Building an Internship Announcement Portal Using Data from www.indeed.com

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Learning Planet Institute - Challeng Hub

Supervisor: Gabrielle FAUSTE | Duration: 4 months

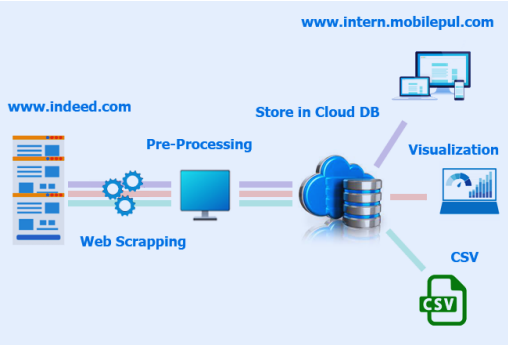
Introduction

This project is about collecting internship related data from indeed.com job announcement website using web scraping method and use collected data to build my own web portal similar to the indeed.com.
This portal will only contain internship announcements located in France.
www.indeed.com is the #1 job site in the world with over 250M unique visitors every month.

Objectives

- Create a web portal which contains only internship related announcements
- Users should be able to search for internships based on a specific location in France
- Learn web scraping and work on real world data than dummy data
- Create my own database environment and interconnect it with Jupyter notebook
- Sharpen my data analysis skills while working with real life project

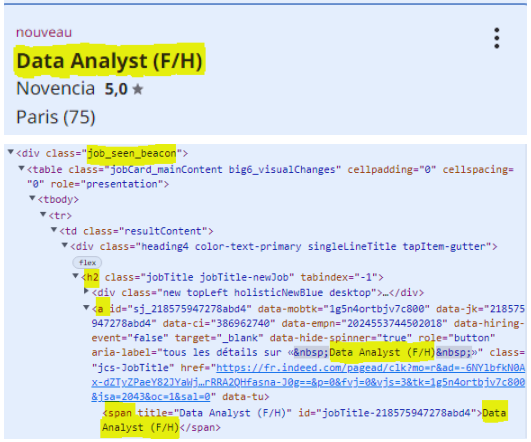
Process Infrastructure



Methodology

Web Scraping:

- HTML format of the page
- Analyze the HTML codes of the page and follow the hierarchical position of required data.
- Write python codes accordingly




```
<div class="job_seen_beacon">
 <table class="jobCard_mainContent big6_visualChanges" cellpadding="0" cellspacing="0" role="presentation">
  <tbody>
   <tr>
    <td class="resultContent">
     <div class="heading4 color-text-primary singleLineTitle tapItem-gutter">
      <h2 class="jobTitle jobTitle-newJob" tabindex="-1">
       <div class="new topLeft holisticNewBlue desktop">…</div>
       <a id="sj_218575947278abd4" data-mobtk="1g5n4ortbjv7c800" data-jk="218575947278abd4" data-ci="386962740" data-empn="2024553744502018" data-hiring-event="false" target="_blank" data-hide-spinner="true" role="button" aria-label="tous les détails sur « Data Analyst (F/H) »" class="jcs-JobTitle" href="https://fr.indeed.com/pagead/clk?mo=r&ad=-6NY1bFkN0Ax-dZTyZPaeY82JYaWj…rRRA2OHfasna-J0g==&p=0&fvj=0&vjs=3&tk=1g5n4ortbjv7c800&jsa=2043&oc=1&sal=0" data-tu>
        <span title="Data Analyst (F/H)" id="jobTitle-218575947278abd4">Data Analyst (F/H)</span>
```

Data Pre-Processing

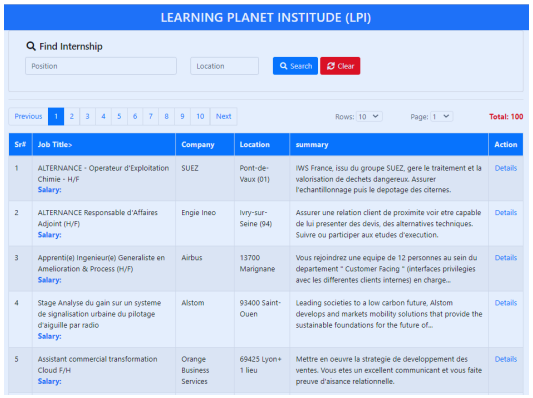
- Remove some HTML related texts
- Remove/Replace symbols character.
- Joint Internship and Apprenticeship announcements
- remove duplicate records
- generate extra columns from existing data before inserting it in to the database

Database Setup and Connection

- Create database on remote/cloud server and set username and password
- Establish connection to the database from jupyter notebook or google colab
- Insert processed data in to the database
- remove duplicate records

Portal Development

For the development of the portal, I have used PHP programming language along with the CSS and HTML. With the help of google, i managed to create my portal with basic function of searching for an specific announcement based on its location.

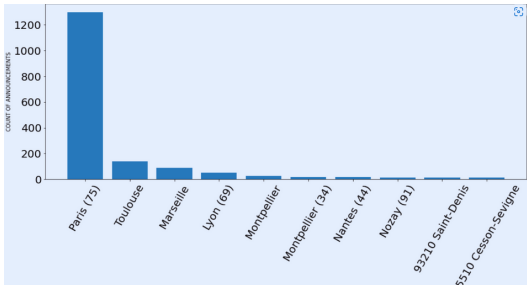


Playing with Data

Setting up Cron Job on the Server



Announcement Count / City



Tools Used

Jupyter, CO, cPanel & WHM, MySQL phpMyAdmin, XAMPP, ClickUp

Challenges Faced

- Collecting data using API was not possible
- Jupyter notebook and remote server interconnection
- Search result limitation on indeed.com to 1000
- Incompatible database for French letters
- Location/posted date comes with different forms

Earned Skills

- Web scrapping (free data collection tool)
- Working with DB on Jupyiter Notebook environment.
- Improved my data analysis skills
- Improved my shell scripting skills